WORKED SOLUTION

STEP II 2004, Question 2

Pure • Quadratic Inequalities

STEP II 2004, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s2-q2

The key idea

The modulus is the only awkward feature, and it disappears if we write $t = |x|$. Since $x^2 = |x|^2$, the inequality becomes an ordinary quadratic inequality in t, with the extra condition $t \geq 0$. Completing the square, or looking at the discriminant, then tells us everything. Each admissible interval of t gives *two* intervals of x, one on each side of the origin.

Solution

Throughout, put $t = |x| \geq 0$. Since $x^2 = t^2$, the inequality $(*)$ says exactly

$$q(t) = t^2 - \alpha t + 2 < 0, \qquad t = |x| \geq 0.$$

No solutions when $|\alpha| < 2\sqrt{2}$

Completing the square,

$$q(t) = \left(t - \frac{\alpha}{2}\right)^2 + 2 - \frac{\alpha^2}{4}.$$

If $|\alpha| < 2\sqrt{2}$ then $\alpha^2 < 8$, so $2 - \frac{\alpha^2}{4} > 0$. The squared term is never negative, so $q(t) > 0$ for *every* real t, and in particular for $t = |x|$. Hence $x^2 - \alpha|x| + 2 > 0$ for all x, and there is no value of x satisfying $(*)$.

The case $\alpha = 3$

Now $q(t) = t^2 - 3t + 2 = (t-1)(t-2)$, which is negative exactly when $1 < t < 2$. So $(*)$ holds if and only if $1 < |x| < 2$, and the solution set is

$$\{x : -2 < x < -1\} \cup \{x : 1 < x < 2\}.$$

The sum of lengths S

Let $\alpha > 2\sqrt{2}$. Then $\alpha^2 > 8$, so q has two distinct real roots

$$t_1 = \frac{\alpha - \sqrt{\alpha^2 - 8}}{2}, \qquad t_2 = \frac{\alpha + \sqrt{\alpha^2 - 8}}{2},$$

and, as q has positive leading coefficient, $q(t) < 0$ exactly when $t_1 < t < t_2$.

Ask yourself Do we have to worry that some of the interval $t_1 < t < t_2$ is at negative t?

No. The roots have product $t_1 t_2 = 2 > 0$ and sum $t_1 + t_2 = \alpha > 0$, so both are positive. Hence the whole interval (t_1, t_2) lies in $t > 0$, and every value of t in it is a genuine value of $|x|$.

So $(*)$ holds if and only if $t_1 < |x| < t_2$, that is,

$$-t_2 < x < -t_1 \qquad \text{or} \qquad t_1 < x < t_2.$$

These are two disjoint intervals (since $t_1 > 0$), each of length $t_2 - t_1 = \sqrt{\alpha^2 - 8}$. Therefore

$$S = 2\sqrt{\alpha^2 - 8}.$$

Since $\alpha^2 - 8 < \alpha^2$ and $\alpha > 0$, taking positive square roots gives $\sqrt{\alpha^2 - 8} < \alpha$, so

$$S < 2\alpha,$$

as required.

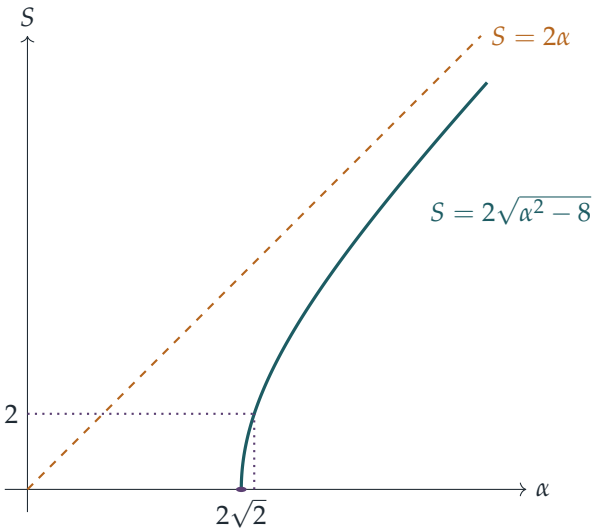
The graph of S against α

The function $S = 2\sqrt{\alpha^2 - 8}$ is defined for $\alpha > 2\sqrt{2}$ (for $\alpha \le 2\sqrt{2}$ there are no intervals at all: at $\alpha = 2\sqrt{2}$ we get $q(t) = (t - \sqrt{2})^2 \ge 0$, and whenever $\alpha \le 0$ we have $q(t) \ge 2$ for all $t \ge 0$). Its main features:

- $S \to 0$ as $\alpha \to 2\sqrt{2}^{+}$, and $\dfrac{\mathrm{d}S}{\mathrm{d}\alpha} = \dfrac{2\alpha}{\sqrt{\alpha^2 - 8}}$, which tends to infinity there: the curve starts at $(2\sqrt{2}, 0)$ with a vertical tangent.
- S is increasing, and $\dfrac{\mathrm{d}^2 S}{\mathrm{d}\alpha^2} = -\dfrac{16}{(\alpha^2 - 8)^{3/2}} < 0$, so the curve is concave.
- The line $S = 2\alpha$ is an asymptote, approached from below: since $(2\alpha - S)(2\alpha + S) = 4\alpha^2 - 4(\alpha^2 - 8) = 32$,

$$2\alpha - S = \frac{32}{2\alpha + S} \to 0 \quad \text{as } \alpha \to \infty.$$

In fact $\frac{\alpha^2}{8} - \frac{S^2}{32} = 1$, so the graph is one quarter of a hyperbola, with asymptote $S = 2\alpha$.



Checks

- At $\alpha = 3$ the formula gives $S = 2\sqrt{1} = 2$, matching the two intervals $(-2, -1)$ and $(1, 2)$ of length 1 each (the dotted point on the graph, at $\alpha = 3$).
- At $\alpha = 5$, $S = 2\sqrt{17} \approx 8.246$; measuring numerically the set of x in $[-20, 20]$ where $x^2 - 5|x| + 2 < 0$ gives the same value, comfortably below $2\alpha = 10$.
- For large α, $2\alpha - S \approx 8/\alpha$: at $\alpha = 100$ the gap is $0.0800\ldots$.

Common slips

- Forgetting that each interval of $|x|$ gives two intervals of x, and so getting $S = \sqrt{\alpha^2 - 8}$.
- Writing the $\alpha = 3$ answer as $1 < x < 2$ only, losing the negative half.
- Using the discriminant without noting that both roots in t are positive; for $\alpha < 0$ the same discriminant condition gives no solutions at all, because the roots are negative.

Beyond STEP

The function $x \mapsto x^2 - \alpha|x| + 2$ is even, so its solution set is symmetric, and the length S is the *measure* of the set where it is negative. Measuring the size of sets defined by inequalities is the starting point of measure theory and Lebesgue integration, met in second-year analysis. For a careful treatment of the limits, derivatives and inequalities used here, read Abbott, *Understanding Analysis*.

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